

Report: Advanced Auditing & Monitoring

1. Introduction

Auditing and monitoring are essential components of a secure and accountable Active Directory environment. This task focused on implementing advanced auditing by configuring System Access Control Lists (SACLs) for privileged AD objects, monitoring key security events in the Event Viewer, and enforcing file-level audit policies on sensitive directories. These actions enable administrators to detect unauthorized privilege changes, object modifications, and file access attempts. All configurations and validations were performed across the domain controller and domain-joined client machines using built-in Windows tools, ensuring comprehensive coverage of auditing requirements.

2. Objective

The objective was to deploy and verify a complete auditing framework for Active Directory and file system objects. This included applying SACLs to critical security groups like Domain Admins, tracking object-level event IDs such as 4732 and 5136, and implementing custom object access auditing for sensitive file locations. The goal was to make all sensitive changes visible and traceable in the Security event logs. By ensuring accurate and consistent audit trails, the environment supports early detection of threats, meets compliance standards, and strengthens internal controls.

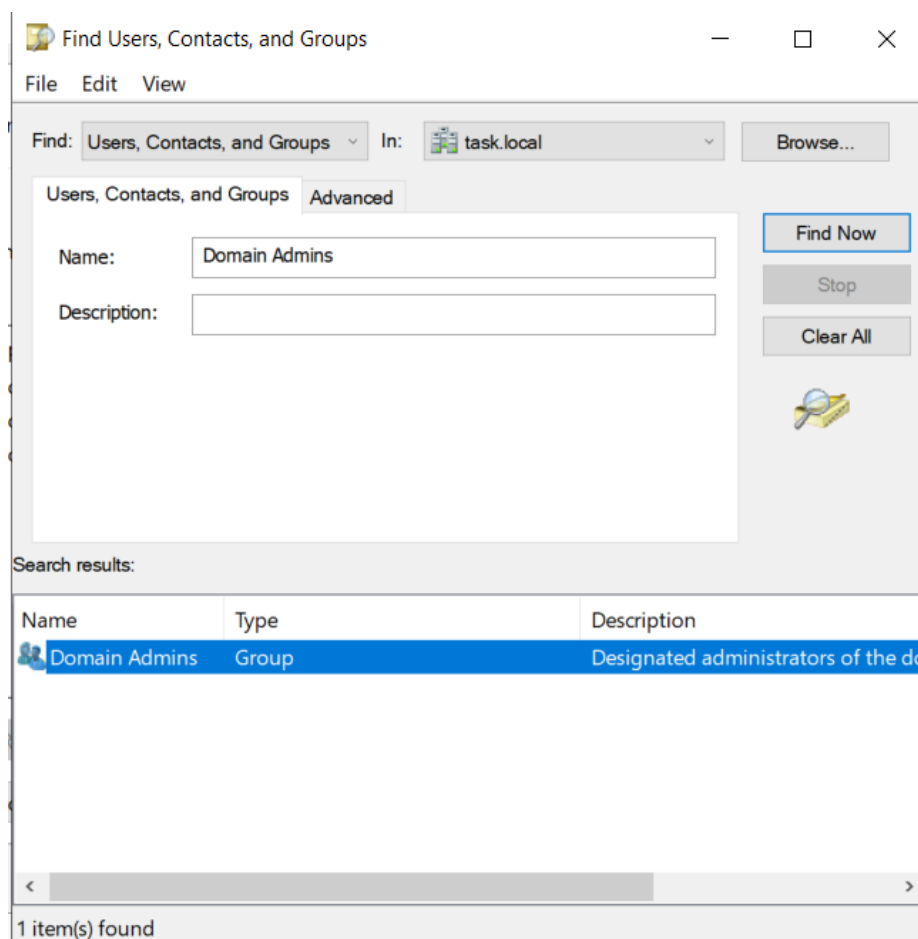
3. Tools used

- Group Policy Management Console (GPMC)
- Active Directory Users and Computers (ADUC)
- Event Viewer
- PowerShell

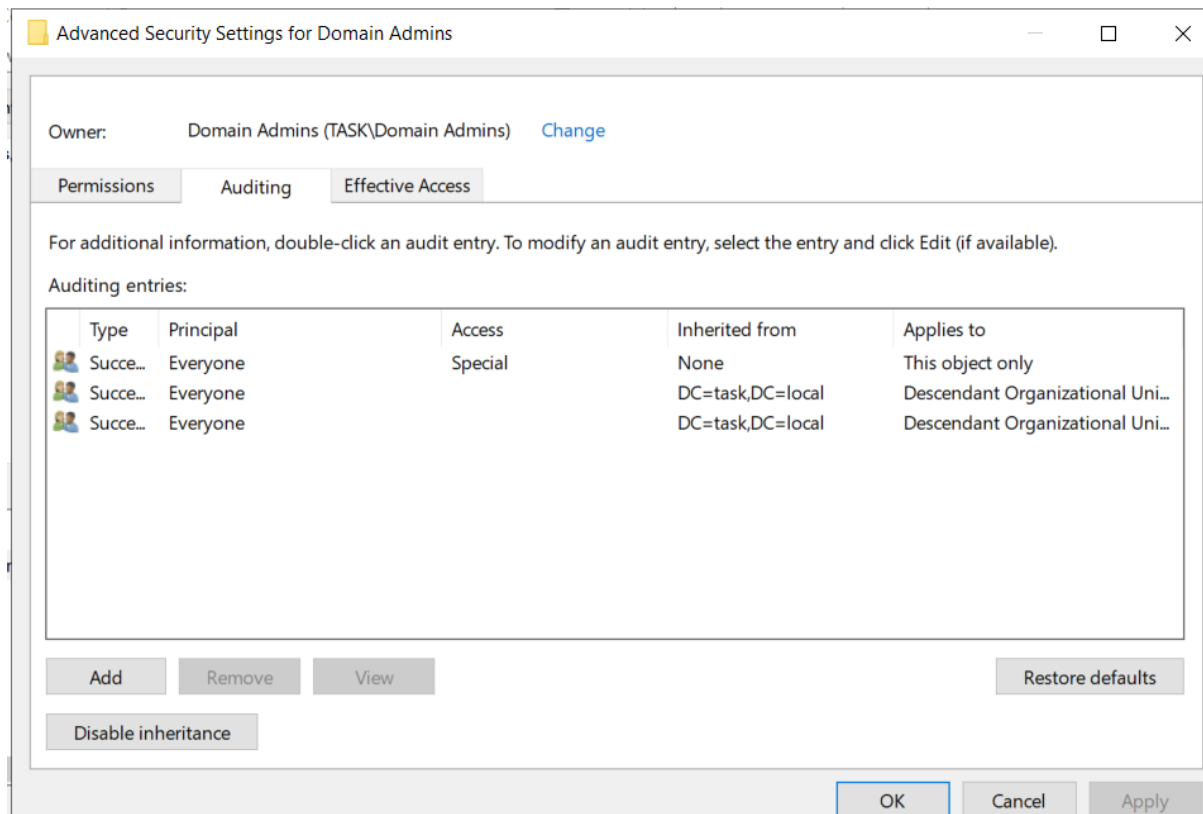
4. Methodology

4.1 Configure SACLs on Critical AD Objects

SACLs were configured on the Domain Admins group by enabling “Advanced Features” in ADUC and accessing the object’s Security settings. Under the Auditing tab, entries were added for the “Everyone” principal with permissions including “Write all properties,” “Modify permissions,” and “Add/remove self as member.” These were scoped to apply to “This object only.” Group Policy was then edited to enable “Audit Directory Service Changes” under Advanced Audit Policy Configuration, and gpupdate /force was used to apply the settings. The applied policies were verified using the auditpol command.



(Find domain admins group)



(Added SACL Entries under Auditing Tab)

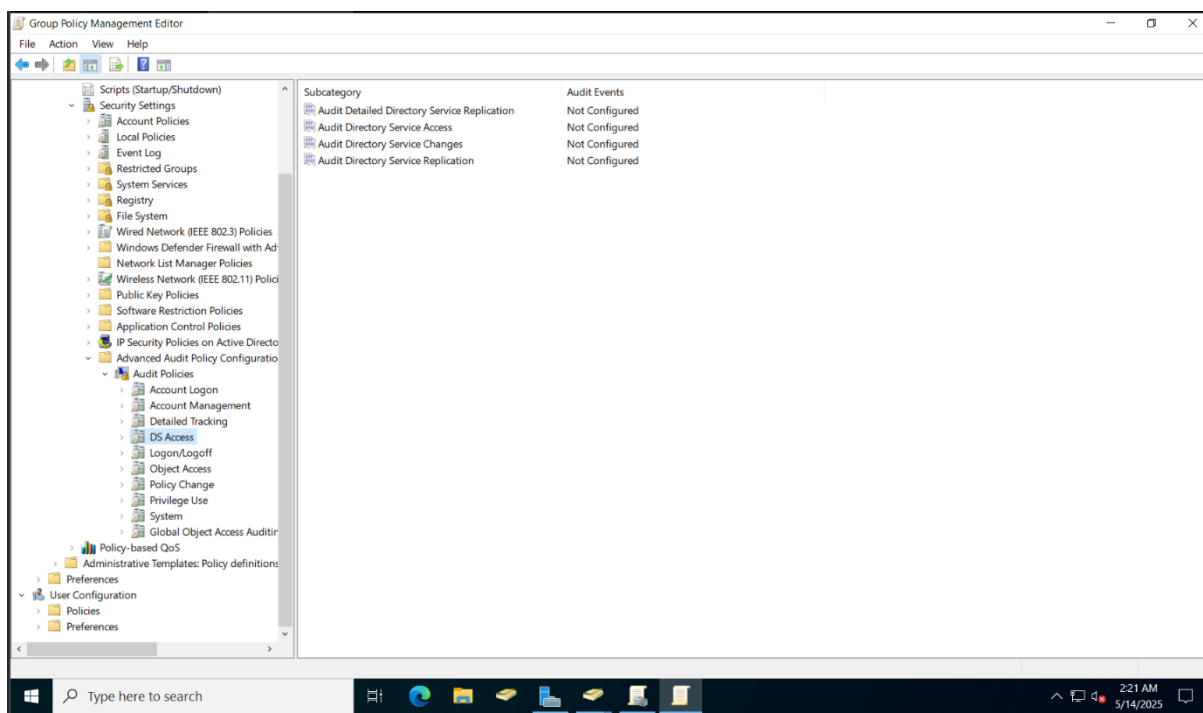
- ☐ Full control
☐ List contents
☐ Read all properties
☒ Write all properties
☐ Delete
☐ Delete subtree
☐ Read permissions
☒ Modify permissions
- ☐ Modify owner
☐ All validated writes
☐ All extended rights
☐ Create all child objects
☐ Delete all child objects
☐ Add/remove self as member
☐ Send to

(Modified settings)

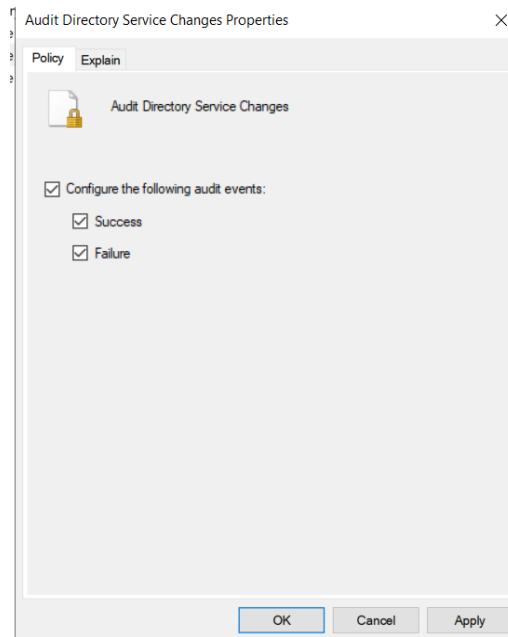
Auditing entries:

Type	Principal	Access	Inherited from	Applies to
All	Everyone	Special	None	This object only
All	Everyone	Add/remove self as me...	None	This object only
Succe...	Everyone		DC=task,DC=local	Descendant Organizational Uni...
Succe...	Everyone		DC=task,DC=local	Descendant Organizational Uni...

(Changes done)



(Now edit default domain policy)



(Edit policy)

```
updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.

PS C:\Users\administrator> gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
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Created on [ 5/14/2025 at 2:24:46 AM

RSOP data for TASK\Administrator on PC1 : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.19044
Site Name:              Default-First-Site-Name
Roaming Profile:        N/A
Local Profile:          C:\Users\administrator
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=PC1,OU=Finance,OU=Departments,DC=task,DC=local
Last time Group Policy was applied: 5/14/2025 at 2:24:21 AM
Group Policy was applied from:      server1.task.local
Group Policy slow link threshold:   500 kbps
Domain Name:                        TASK
Domain Type:                        Windows 2008 or later

Applied Group Policy Objects
-----
  Disable_USB_Finance
  Logon_Audit_Policy
  Object_Access_Audit
  ScreenLock_15Min
  Disable_SMBv1
  Logon_Banner_GPO
  Disable_NTLM
  LAPS_Policy
  Default Domain Policy
```

(We made changes in already existing logon so it's already here)

```
PS C:\Users\Administrator> auditpol /get /category:"DS Access"
System audit policy
Category/Subcategory          Setting
DS Access
  Directory Service Changes    Success and Failure
  Directory Service Replication No Auditing
  Detailed Directory Service Replication No Auditing
  Directory Service Access      No Auditing
PS C:\Users\Administrator> _
```

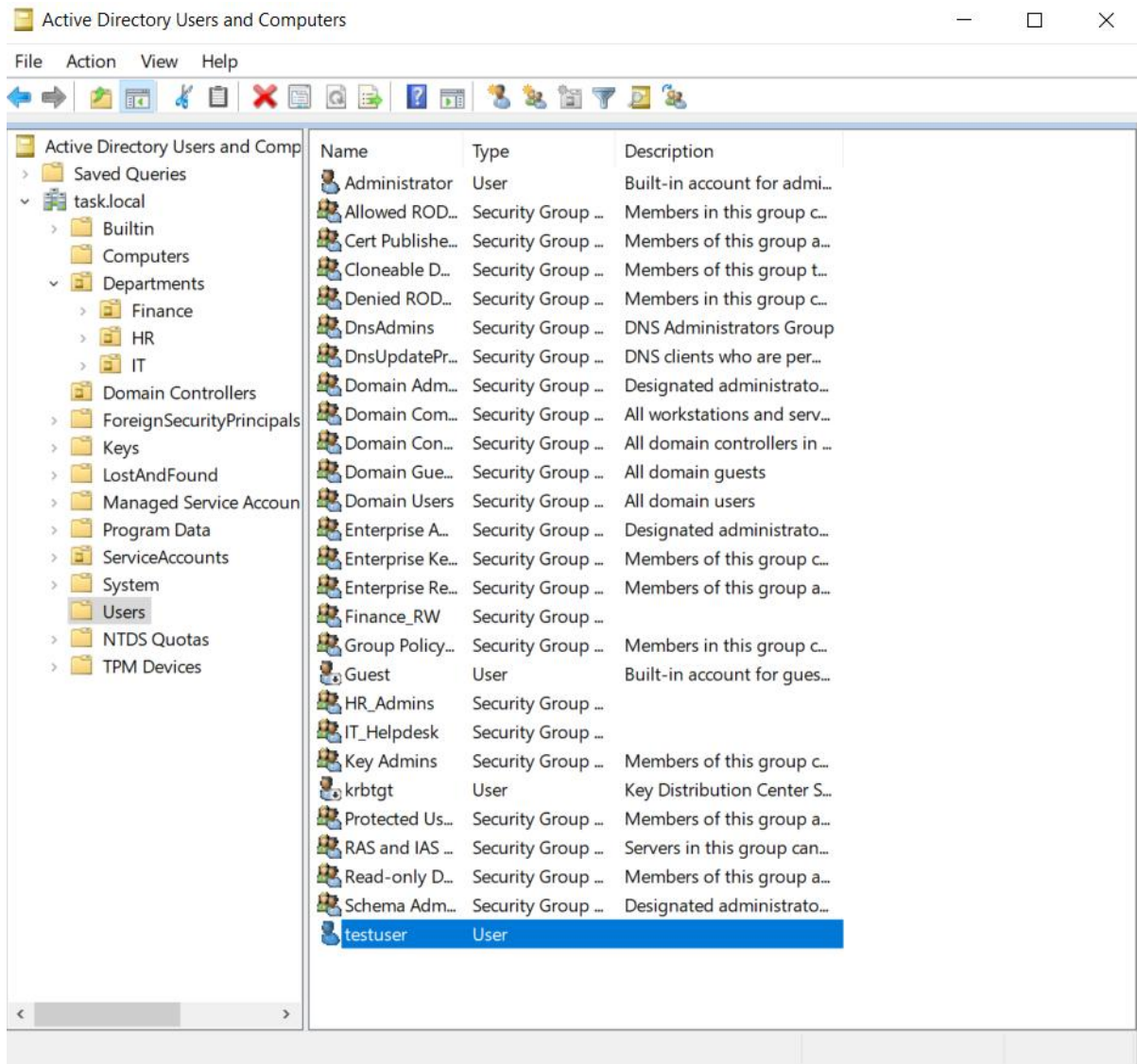
(Verified)

4.2 Monitor Event Logs for Unauthorized Changes

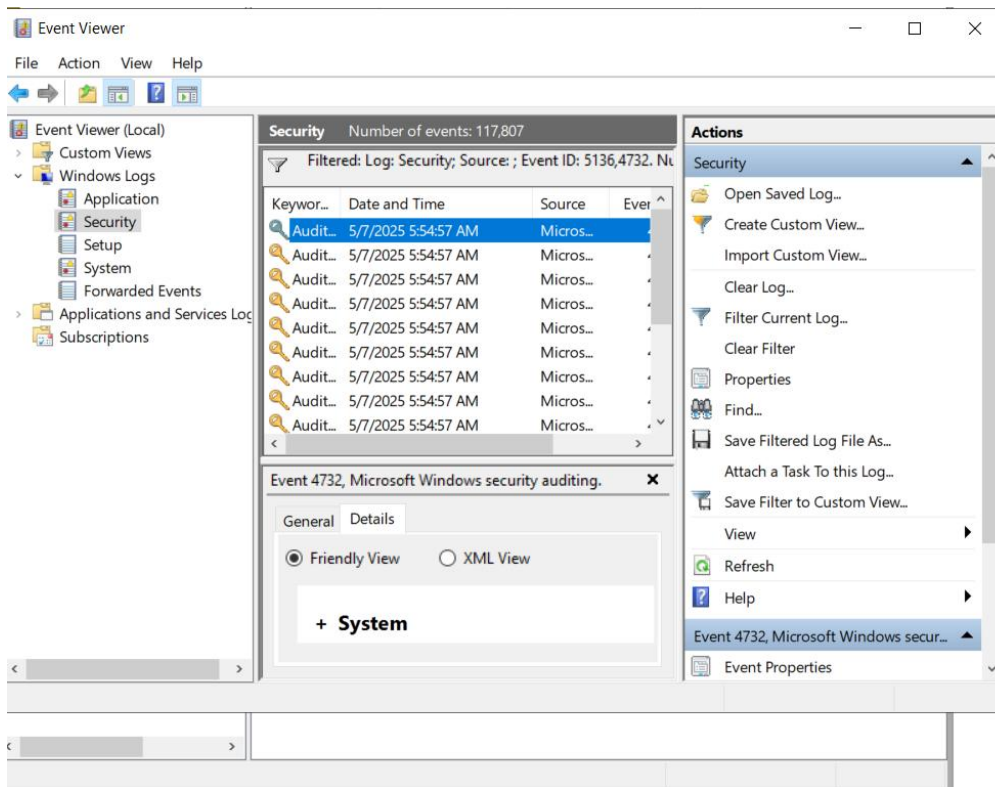
Changes were simulated by adding a user to the Domain Admins group. The Event Viewer on the domain controller was used to review logs under the Security category. Filtering for Event IDs 4732 and 5136 successfully revealed:

- 4732: A member was added to a security-enabled global group
- 5136: A directory service object was modified

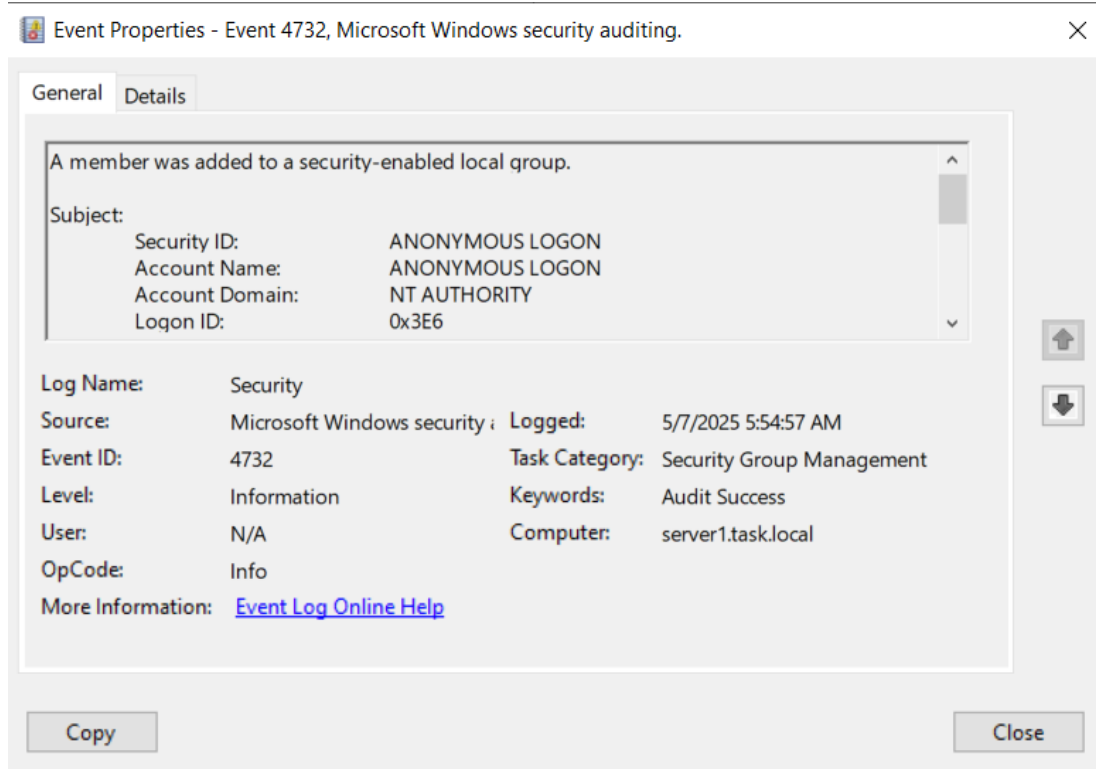
These events confirmed that sensitive changes were correctly captured and attributed to the responsible user accounts.



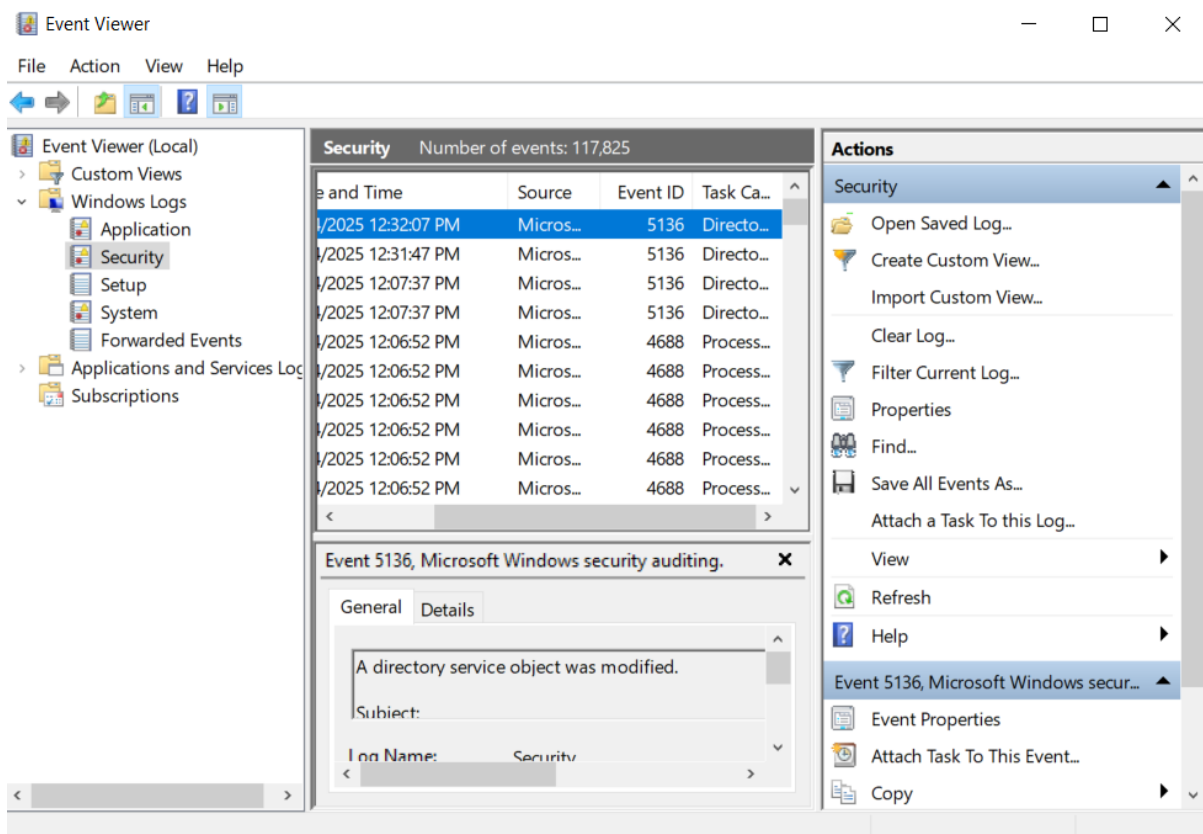
(Added a new user-testuser)



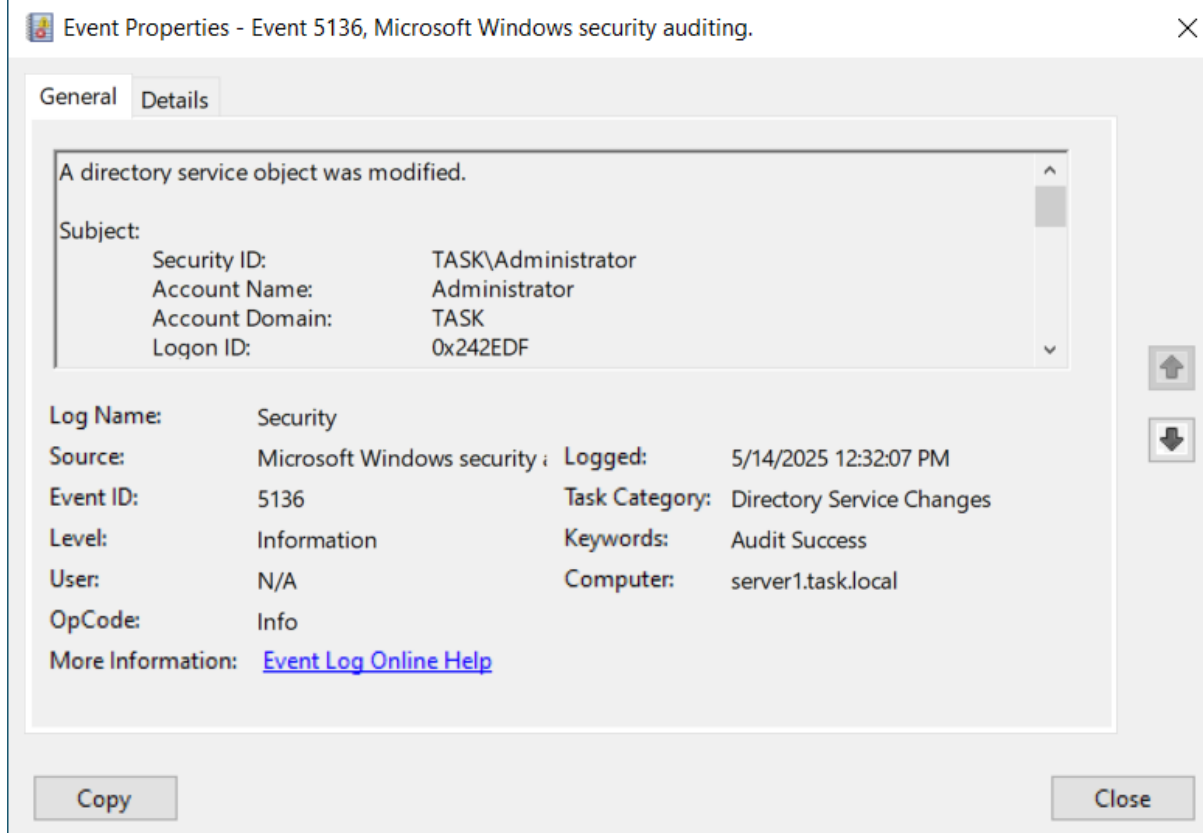
(Open event viewer)



(Log for 4732)



(Log for 5136)

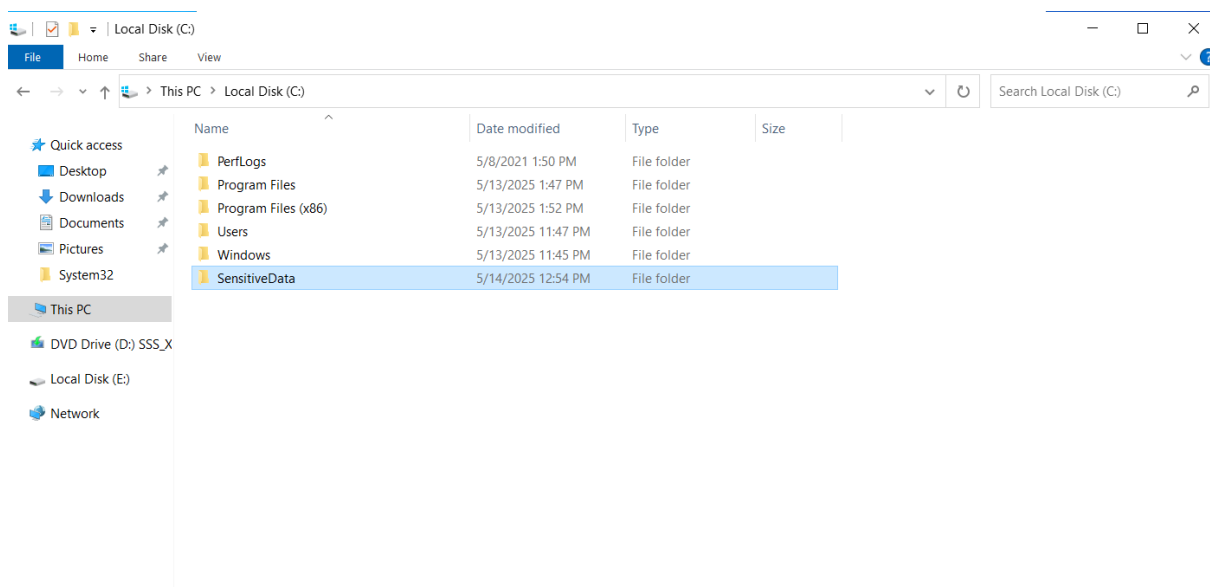


(Event log)

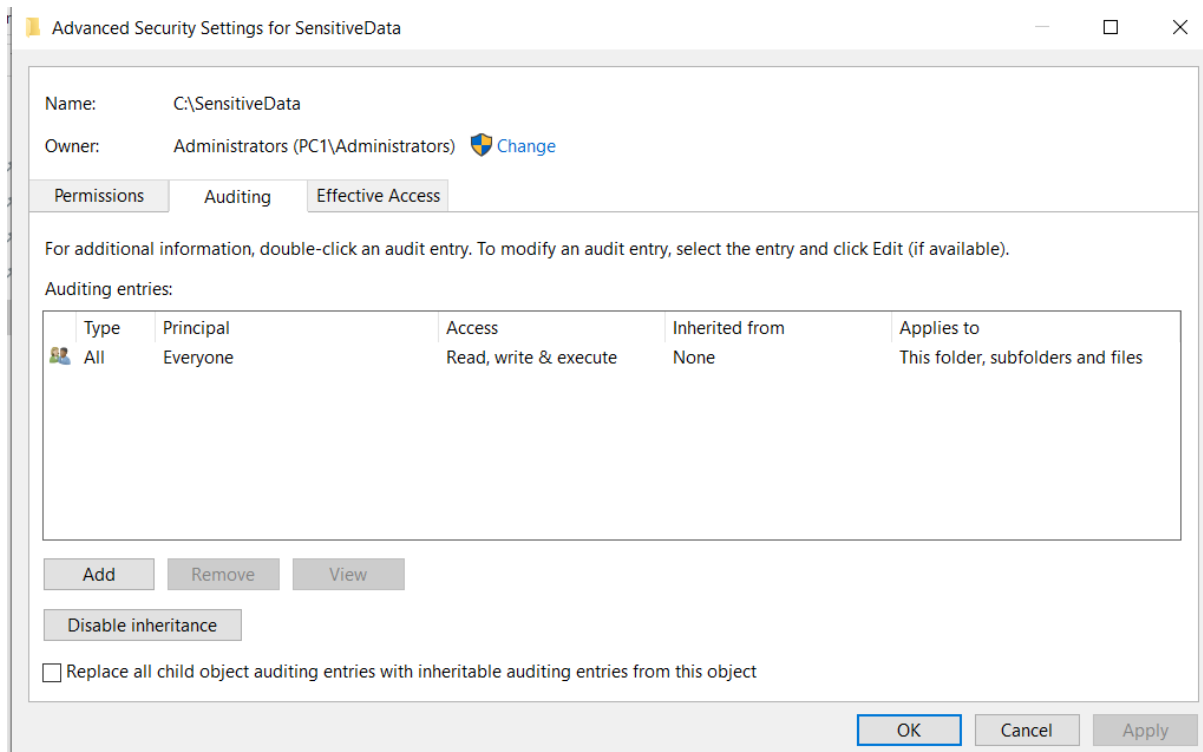
NOTE: For log id 5136, added an another testuser in Admin Domains and then deleted it.

4.3 Set Up Custom Audit Policy for Sensitive File Access

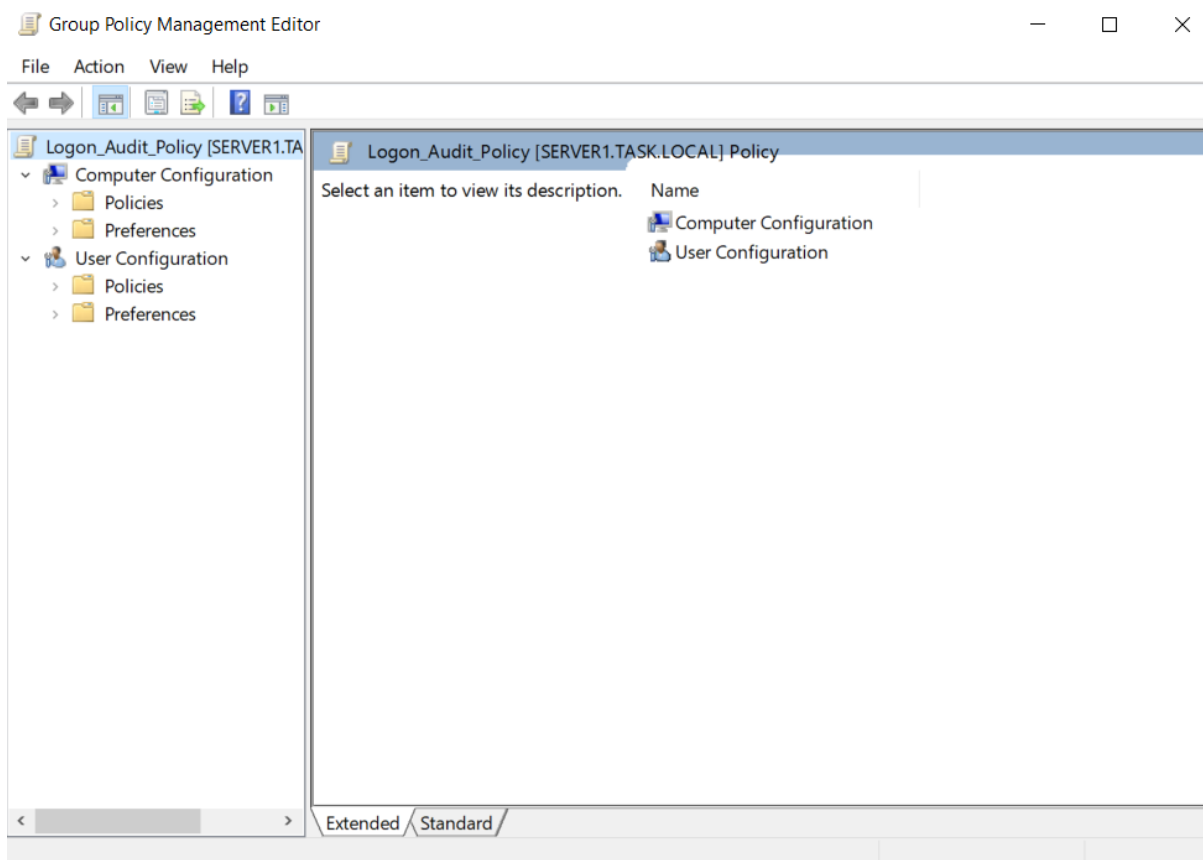
A test folder named C:\SensitiveData was created on the domain-joined client machine. The folder's Auditing settings were configured to log read and write access for all users. In Group Policy, the "Audit File System" option was enabled under Object Access for Success and Failure. After forcing a GPO update, actions such as reading or writing to the folder as the aarav user triggered Event ID 4663 in the client's Event Viewer, showing the accessed object, user account, and access type. This confirmed the effectiveness of the custom file audit policy.



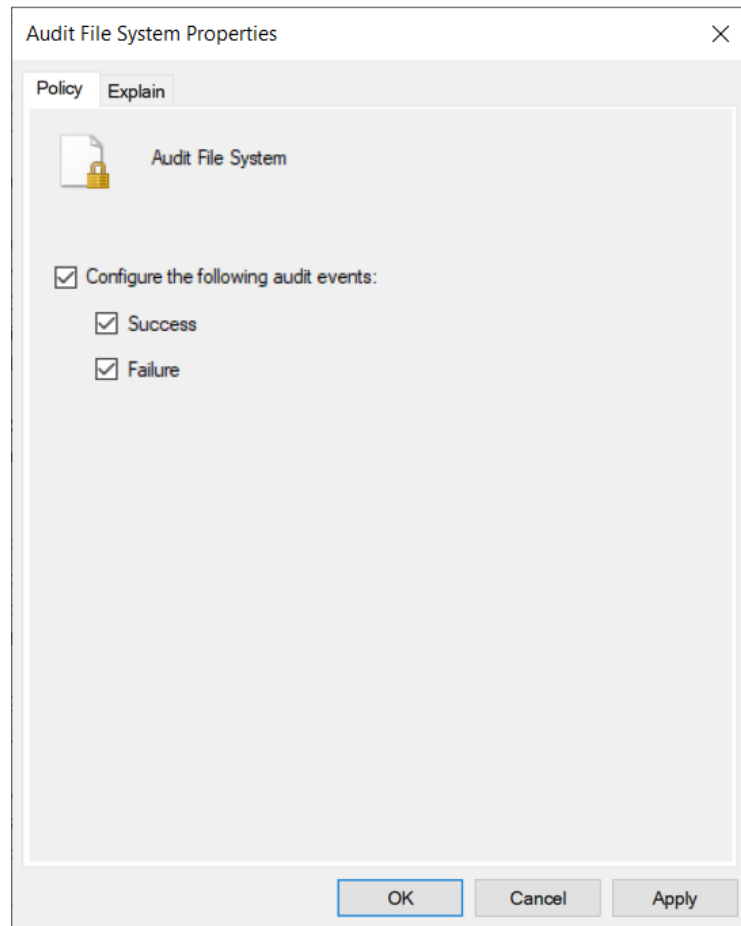
(Create a new folder SensitiveData in PC1 in admin account)



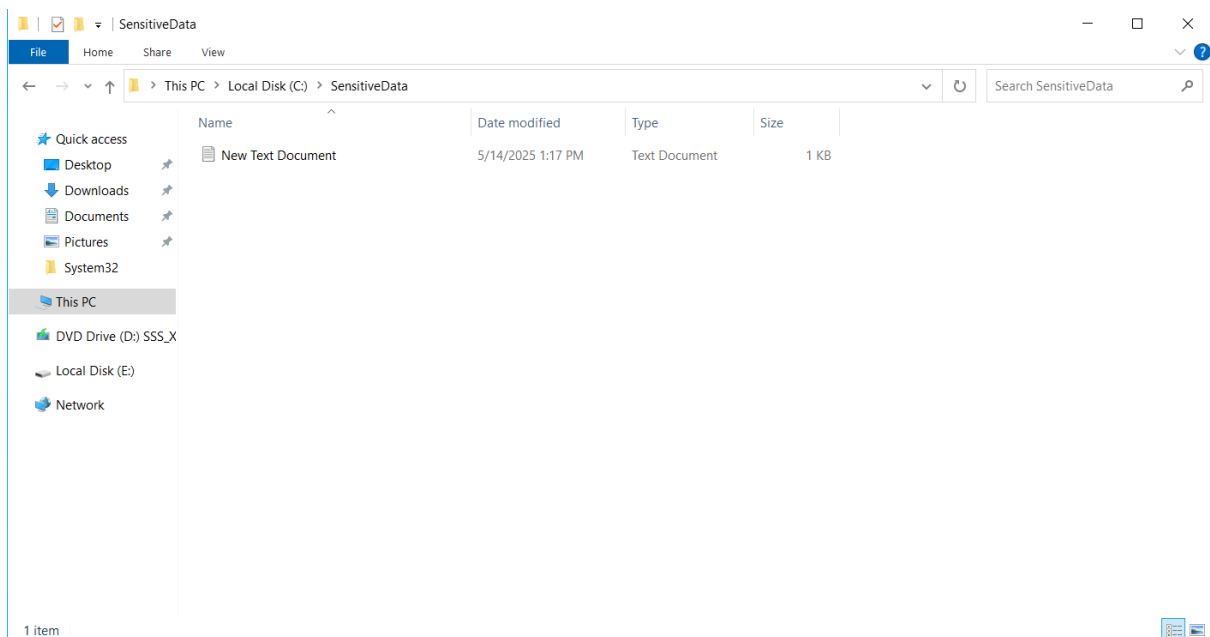
(Edit auditing policy for this folder)



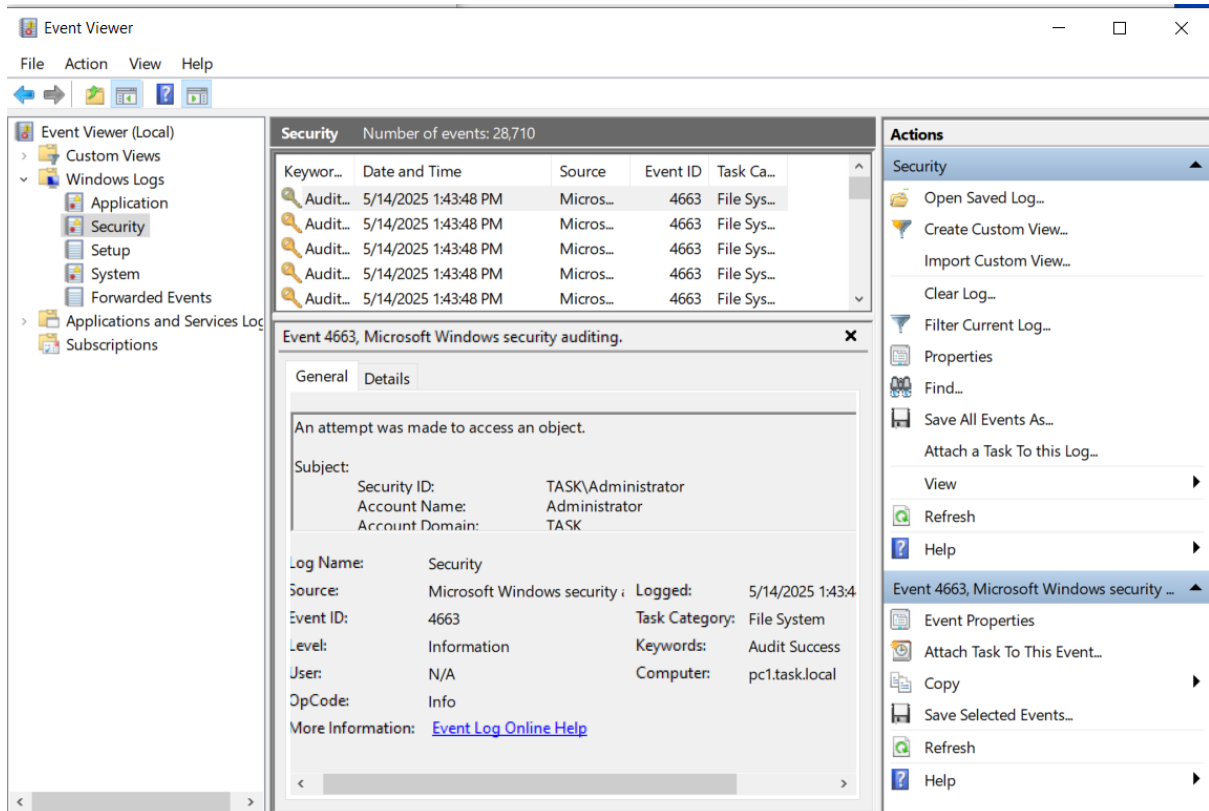
(Edit Logon_Audit_Policy which we used in previous task)



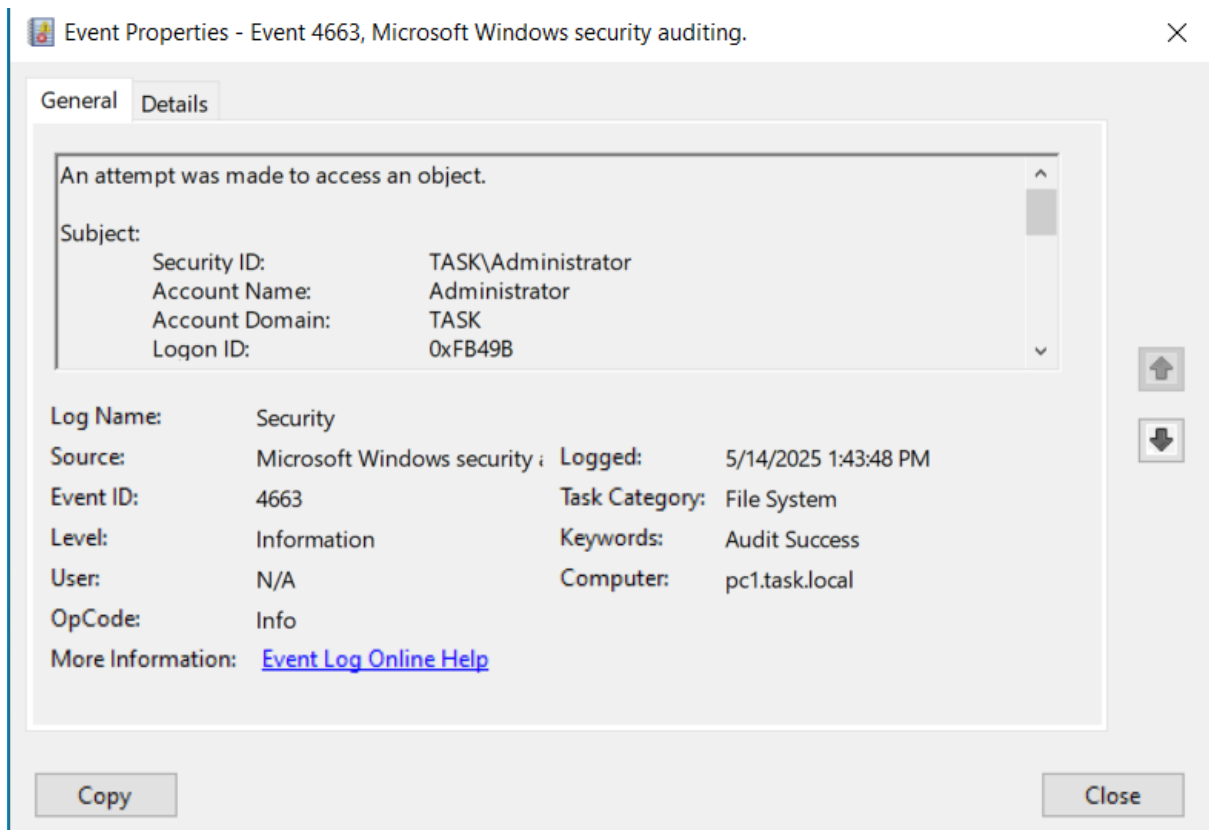
(Edit policy of advance auditing policy)



(make a new file in SensitiveData folder in PC1 in user account)



(We can see event id 4663 in event viewer)



(Event details)

5. Recommendations

- Apply SACLs only to high-value AD objects to minimize log noise.
- Regularly monitor Security logs for event IDs 4732, 5136, and 4663.
- Ensure only authorized personnel can modify SACL and audit policy settings.
- Use Group Policy to enforce consistent auditing across all domain machines.
- Implement log forwarding or SIEM integration for centralized visibility.
- Periodically review and archive audit logs for compliance and forensic readiness.

6. Conclusion

This task established a reliable auditing framework for tracking critical Active Directory and file-level activities. By applying SACLs to the Domain Admins group and enabling detailed audit policies via GPO, the system now logs sensitive actions such as group membership changes and file access attempts. The corresponding event IDs (4732, 5136, and 4663) were verified in the Security logs, validating that the auditing configurations are functional and effective. These controls improve the organization's ability to detect, investigate, and respond to unauthorized behavior, thereby strengthening the overall security posture.